

School of Enterprise Computing & Digital Transformation,
TU Dublin (Tallaght)
M.Sc. in DevOps
2025/2026

Enterprise Architecture Deployment

CA2 Project
Value: 70%

Lecturer: Dr Omar Portillo

Overview:

In previous modules, you have deployed a web application in a CI/CD Pipeline using Microsoft Azure's product suite. However, this has been done in the same manner as a monolithic application would. The application contained tightly coupled user-facing and business logic, with possibly no independent data layer.

DevOps is experiencing a paradigm shift, and many enterprises are now drawn to breaking their monolithic systems down into the microservice architectural style. The goal of this assignment is to fully investigate and implement an enterprise-style CI/CD deployment based on the **microservice** architectural style, which contains at least **two** services and **one** data layer.

There are two components to this assignment: [1] your investigation (along with its companion report); [2] a loosely coupled service-oriented application that is deployed using some (or all) of the components from your research. It must be deployed in a CI/CD pipeline that is provisioned in a truly automated manner; [3] a demonstration/presentation of the previous two components.

Microservice Deployment:

You may deploy your own service-oriented application in a language (as well as business functionality) of your choice, but the overall architecture must include at least two communicating services and a data layer (database or file system). Alternatively, you will be provided with some simple services you can use (to build on, if you wish) to create and manage your CI/CD deployment. The architecture of the provided sample services includes a dynamic front-end (FE) service that communicates with a Web Service back-end (BE), and a database. The functionality offered by these services is a (very basic/rudimentary!) recipe management web application. More specifically, this application includes a RESTful Java web service BE, a Node.js FE service, and a MongoDB database (to store recipes) that communicate with each other.

Research Details:

You are required to produce a detailed **Release Management Plan (RMP)** for your enterprise-style CI/CD deployment. This will entail researching and evaluating options for developing your Continuous Integration/Test/Deployment RMP. The tool choice should be secondary. That is, select tools that are

suitable for the work required. Do not decide how to work based on the tools.

There is a requirement to fully automate (i.e., IaC) the provisioning and release orchestration of your application. This means your provisioning and release orchestration needs to include a well-thought-out automated release management plan that incorporates automated infrastructure delivery and continued configuration management. This will ensure a full realisation of the promise of Continuous Delivery. Until the processes associated with a release are fully automated, the pipeline continues to rely on manual processes that can lead to configuration drift.

There is a requirement to have a consistent and repeatable change management process in place. Your deployment should be able to destroy and replace your container orchestration system in a reliable, automated manner. For example, if you use Terraform to provision your container orchestration architecture and its associated services, how are subsequent updates deployed? How can this be achieved while maintaining, for example, a blue/green deployment?

Some decisions to be made are: which deployment strategy will you follow?; should each service have its own code repository?; should each service have its own CI/CD pipeline? How can you make all versions (and tools) unified? How can you make your builds run in the same automated way? What about your backup and recovery strategy?

Research Report:

Produce a report for your **Release Management Plan**. It should detail the specifications and justifications for your deployment strategies. You must evaluate and critique this under a set of criteria which should include (*but not be limited to*):

- Performance.
- Ease of configuration/installation.
- Cost and licensing.
- Monitoring and logging.
- Scaling (in and out).
- Rollback plan.
- Backup and restore strategy.
- Security.
- Support.
- Vulnerability checks on images.
- Sustainability.

Deliverables:

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| 1. Fully automated CI/CD pipeline. | 45 Marks |
| 2. Demonstration of pipeline and deployed services. | 10 Marks |
| 3. Written release management plan (RMP) report. | 35 Marks |
| • Includes a presentation showcasing the highlights of the RMP. | |
| 4. Additional features (at least 2). | 10 Marks |